

MICHAEL C. PUENTE

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MECHANICAL ENGINEERING | PRECISION MANUFACTURING | GEAR DESIGN | AEROSPACE

ENGINEERING PROFILE

Mechanical Engineering student seeking a hands-on manufacturing engineering opportunity with Rave Gears. Experience includes spur-gear transmission and shaft design, AGMA analysis, SolidWorks assemblies and drawings, technical drawing review, equipment verification, and aerospace CFD research. Eager to develop expertise in precision CNC machining, gear manufacturing, inspection, and continuous improvement.

EDUCATION

The University of Texas at San Antonio (UTSA) - B.S. Mechanical Engineering

Expected Fall 2027

Relevant Coursework: Manufacturing Engineering, Machine Element Design, Measurements and Instrumentation, Mechanics of Solids, Numerical Methods, Dynamic Systems and Controls

Texas Lutheran University - Applied Physics Coursework

Activities: TLU Varsity Football | American Society of Mechanical Engineers | Society of Physics Students

TECHNICAL SKILLS

Gear and Mechanical Design: SolidWorks, AGMA gear analysis, shaft/bearing/key design, V-M-T loading, Modified Goodman fatigue analysis, MATLAB FEM

Manufacturing and Quality: Engineering drawing and specification review, equipment verification, measurement and instrumentation, calipers, multimeters, hand and power tools

Engineering Software: MATLAB/Simulink, OpenFOAM, ParaView/ParaFoam, Linux, Excel, Bluebeam Revu, AutoCAD/Revit, Python/Jupyter

SELECTED ENGINEERING PROJECTS

Two-Stage Spur-Gear Speed Reducer - Machine Element Design

SolidWorks | MATLAB

- Designed an 82-hp, two-stage spur-gear transmission reducing 1,200 rpm to 100 rpm through 4:1 and 3:1 stages for an overall 12:1 reduction.
- Performed AGMA bending/contact analysis, V-M-T loading, Modified Goodman fatigue analysis, MATLAB beam-element FEM, and bearing and key/keyway selection.
- Created a complete SolidWorks assembly and dimensioned drawings for input, intermediate, and output shafts, integrating stepped shaft geometry, gears, bearings, and keys.

Jet Engine CAD and Heat Exchanger Flow Simulation

SolidWorks

- Built multi-component jet-engine and heat-exchanger assemblies; modified a heat-exchanger reference design and used Flow Simulation to evaluate outlet temperature, flow paths, and temperature contours.

1994 Ford F-150 Mechanical Restoration

Hands-On Project

- Restored a non-running truck through systematic fuel, ignition, starting, electrical, and mechanical diagnostics, troubleshooting, and component replacement.

ENGINEERING AND TECHNICAL EXPERIENCE

Undergraduate Researcher - Klesse Undergraduate Research Program, UTSA

Summer 2026

- Conducted OpenFOAM simulations of Mach 6 turbulent boundary-layer flow over adiabatic flat-plate and three-dimensional cavity geometries; evaluated pressure, skin friction, thermal behavior, and flow fields.
- Built meshes and boundary conditions, executed parallel jobs on TACC and ARC HPC systems, and interpreted results in ParaView/ParaFoam.

Mechanical Engineering Intern - Sur Texas LLC, San Antonio, TX

Jan. 2025 - Present

- Review commercial HVAC/plumbing bid drawings and perform quantity takeoffs for ductwork, piping, equipment, fixtures, and accessories using Bluebeam Revu and Excel.
- Track revisions, scope changes, and bid updates to improve material coverage, technical clarity, and estimating accuracy across commercial projects.

Warehouse Associate - Texas Air Products, San Antonio, TX

Jul. 2026 - Present

- Receive, verify, identify by tag and application, stage, and load commercial HVAC equipment and components for contractor orders.
- Develop hands-on equipment knowledge while supporting contractor needs across dampers, air-distribution devices, controls, VFDs/starters, curbs, isolators, ductwork, and accessories.